

```
*****
*****
*****Author: Lina Hedman [lina.hedman@ibf.uu.se]
*****Project: Siblings Neighbourhood Effects
*****
*****
```

```
*****
*****MASTER*****
*****
```

```
clear
version 14.0
capture log close
set scheme s1manual
set more off
```

```
*****
*****Global paths*****
*****
```

```
global path "C:\Users\LinaH\Documents\siblings nbd effects\
```

```
global logs "$path\logs"
global output "$path\output"
```

```
log using "$logs\$$_DATE.log", append
```

```
*****
*****Generate population, merge basic datasets and create variables*****
*****
```

```
do "$path\Slave1_createdataset.do"
```

```
*****
*****Generate neighbourhood data*****
*****
```

```
do "$path\Slave2_nbd.do"
```

```
*****
*****Create random sibling pairs and merge with original file*****
*****
```

```
do "$path\Slave3_randompairs.do"
```

```
*****
*****Analysis*****
*****
```

```
do "$path\Slave4_analysis.do"
```

```

*****
*****
*****Author: Lina Hedman [lina.hedman@ibf.uu.se]
*****Project: Siblings Neighbourhood Effects
*****
*****S L A V E (master: MASTER.do)*****
*****

```

```

*****
*****Slave1_createdataset.do
*****

```

```

*****
*****Identify individuals*****
*****

```

```

*Select individuals of age range 15-21 in 1990
use "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\IBF_IndData1990_2014.dta", clear
gen age1990 = 1990-FoddAr
keep if age1990 >= 15 & age1990 <= 21
rename Kon sex
keep IBFLopNr IBFLopNrMor IBFLopNrFar sex FoddAr

```

*Add info on FamID and FamID of parents to identify when the child leaves the parental home

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000{
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(FamLopNr)
drop if _merge == 2
drop _merge
rename FamLopNr FamLopNr`i'
}

```

```

foreach i of numlist 2001 2003 2004 2005 2006 2007 2008 2009 2010{
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(FamLopNr)
drop if _merge == 2
drop _merge
rename FamLopNr FamLopNr`i'
}

```

```

rename IBFLopNr IBFLopNr_child
rename IBFLopNrMor IBFLopNr

```

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 {
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(FamLopNr)
drop if _merge == 2
drop _merge
rename FamLopNr FamLopNrMor`i'
}

```

```

}
rename IBFLopNr IBFLopNrMor
rename IBFLopNrFar IBFLopNr

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 {
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(FamLopNr)
drop if _merge == 2
drop _merge
rename FamLopNr FamLopNrFar`i'
}
rename IBFLopNr IBFLopNrFar
rename IBFLopNr_child IBFLopNr

```

```

*live home in 1990 - keep only the ones who do
gen livewpar90 = 1 if FamLopNr1990 == FamLopNrMor1990 | FamLopNr1990 == FamLopNrFar1990
keep if livewpar90 == 1
drop livewpar90

```

```

*year of leaving parental home - keep only those who move out between 1991 and 1997
gen yrleavehome = .
replace yrleavehome = 1991 if FamLopNr1990 == FamLopNrMor1990 | FamLopNr1990 ==
FamLopNr1990_far & FamLopNr1990 != .
replace yrleavehome = 1992 if FamLopNr1991 == FamLopNrMor1991 | FamLopNr1991 ==
FamLopNr1991_far & FamLopNr1991 != .
replace yrleavehome = 1993 if FamLopNr1992 == FamLopNrMor1992 | FamLopNr1992 ==
FamLopNr1992_far & FamLopNr1992 != .
replace yrleavehome = 1994 if FamLopNr1993 == FamLopNrMor1993 | FamLopNr1993 ==
FamLopNr1993_far & FamLopNr1993 != .
replace yrleavehome = 1995 if FamLopNr1994 == FamLopNrMor1994 | FamLopNr1994 ==
FamLopNr1994_far & FamLopNr1994 != .
replace yrleavehome = 1996 if FamLopNr1995 == FamLopNrMor1995 | FamLopNr1995 ==
FamLopNr1995_far & FamLopNr1995 != .
replace yrleavehome = 1997 if FamLopNr1996 == FamLopNrMor1996 | FamLopNr1996 ==
FamLopNr1996_far & FamLopNr1996 != .
replace yrleavehome = 1998 if FamLopNr1997 == FamLopNrMor1997 | FamLopNr1997 ==
FamLopNr1997_far & FamLopNr1997 != .
replace yrleavehome = . if yrleavehome == 1998 & FamLopNr1997 == .
replace yrleavehome = . if FamLopNr1990 == . & FamLopNr1991 == . & FamLopNr1992 == . &
FamLopNr1993 == . & FamLopNr1994 == . & FamLopNr1995 == . & FamLopNr1996 == . &
FamLopNr1997 == .
label variable yrleavehome "first year living away from parents"

```

```

keep if yrleavehome >= 1991 & yrleavehome <= 1997

```

```

gen yrleave_14 = yrleavehome+13
label variable yrleave_14 "year 14 years after having left parental home"

```

```

gen whichparent = 1 if FamLopNr1990 == FamLopNrMor1990
replace whichparent = 2 if FamLopNr1990 == FamLopNrFar1990
replace whichparent = 3 if FamLopNr1990 == FamLopNrMor1990 & FamLopNr1990 ==
FamLopNrFar1990

```

```
label variable whichparent "which parent ego lives with before move"
label define whichlabs 1 "mother" 2 "father" 3 "both"
label values whichparent whichlabs
```

```
drop FamLopNr# FamLopNrMor# FamLopNrFar#
```

```
/* Save separate files for girls and boy to create same-sex sibling pairs
save "$path\girls.dta"
```

```
keep if sex == 1
save "$path\boys.dta"
```

```
use "$path\girls.dta"
keep if sex == 2
save "$path\girls.dta", replace
```

```
*****
*****Create siblings*****
*****
```

```
*****Female sibling pairs*****
*****
```

```
use "$path\girls.dta", clear
bysort IBFLopNrMor IBFLopNrFar: gen sib_n = _n
drop if sib_n < 2
```

```
reshape wide IBFLopNr sex FoddAr yrleavehome yrleave_14 whichparent, i(IBFLopNrMor
IBFLopNrFar) j(sib_n)
gen sibID = _n
reshape long sibID IBFLopNr sex FoddAr yrleavehome yrleave_14 whichparent sib_n, i(IBFLopNrMor
IBFLopNrFar) j(sibID_within)
drop if IBFLopNr == .
```

```
*Keep siblings in right age range
sort sibID FoddAr
gen age_diff =(FoddAr - FoddAr[_n+1] if sibID == sibID[_n+1])
replace age_diff = abs(age_diff)
bysort sibID: egen age_diff_small = min(age_diff)
keep if age_diff_small <=3
```

```
* Keep families where at least one child leaves home between 1991 and 1993
gen home91_93 = 1 if yrleavehome >=1991 & yrleavehome <= 1993
bysort sibID: egen sibhome91_93 = count(home91_93)
keep if sibhome91_93 >= 1
```

```
*Keep only siblings who leave home min 4 years apart
sort sibID yrleavehome
gen leave_diff =(yrleavehome - yrleavehome[_n+1] if sibID == sibID[_n+1])
```

```

replace leave_diff = abs(leave_diff)
bysort sibID: egen leave_diff_small = min(leave_diff)
keep if leave_diff_small <=4

*Keep only pair closest in age
gen sibpair = 1 if age_diff == age_diff_small
replace sibpair = 1 if age_diff[_n-1] == age_diff_small[_n-1]
keep if sibpair == 1

*keep only siblings living with same parent
bysort sibID: gen samepar = 1 if whichparent == whichparent[_n+1]
replace samepar = 1 if whichparent == whichparent[_n-1] & sibID == sibID[_n-1]

*Drop created variables
drop sibID_within sib_n age_diff age_diff_small home91_93 sibhome91_93 leave_diff
leave_diff_small sibpair samepar

*generate sibID different from males(avoid replicas)
replace sibID = 2000000+sibID

*gen sibling sex variable
gen sib_sex == 2
label define sib_sex_label 1 "male siblings" 2 "female siblings"
label values sib_sex sib_sex_label

save "$path\girls.dta", replace

*****
*****Male sibling
pairs*****
*****

use "$path\boys.dta", clear
bysort IBFLopNrMor IBFLopNrFar: gen sib_n = _n
drop if sib_n < 2

reshape wide IBFLopNr sex FoddAr yrleavehome yrleave_14 whichparent, i(IBFLopNrMor
IBFLopNrFar) j(sib_n)
gen sibID = _n
reshape long sibID IBFLopNr sex FoddAr yrleavehome yrleave_14 whichparent sib_n, i(IBFLopNrMor
IBFLopNrFar) j(sibID_within)
drop if IBFLopNr == .

*Keep siblings in right age range
sort sibID FoddAr
gen age_diff =(FoddAr - FoddAr[_n+1] if sibID == sibID[_n+1])
replace age_diff = abs(age_diff)
bysort sibID: egen age_diff_small = min(age_diff)
keep if age_diff_small <=3

* Keep families where at least one child leaves home between 1991 and 1993
gen home91_93 = 1 if yrleavehome >=1991 & yrleavehome <= 1993

```

```
bysort sibID: egen sibhome91_93 = count(home91_93)
keep if sibhome91_93 >= 1
```

```
*Keep only siblings who leave home min 4 years apart
sort sibID yrleavehome
gen leave_diff =(yrleavehome - yrleavehome[_n+1] if sibID == sibID[_n+1])
replace leave_diff = abs(leave_diff)
bysort sibID: egen leave_diff_small = min(leave_diff)
keep if leave_diff_small <=4
```

```
*Keep only pair closest in age
gen sibpair = 1 if age_diff == age_diff_small
replace sibpair = 1 if age_diff[_n-1] == age_diff_small[_n-1]
keep if sibpair == 1
```

```
*keep only siblings living with same parent
bysort sibID: gen samepar = 1 if whichparent == whichparent[_n+1]
replace samepar = 1 if whichparent == whichparent[_n-1] & sibID == sibID[_n-1]
```

```
*Drop created variables
drop sibID_within sib_n age_diff age_diff_small home91_93 sibhome91_93 leave_diff
leave_diff_small sibpair samepar
```

```
*generate sibID different from females(avoid replicas)
replace sibID = 1000000+sibID
```

```
*gen sibling sex variable
gen sib_sex == 1
label values sib_sex sib_sex_label
```

```
save "$path\boys.dta", replace
```

```
*****
*****Merge boys and girls into one file containing full sibling pair population*****
*****
```

```
append using "$path\girls.dta"
gen siblingtype = 1
label variable siblingtype "real or random pair"
label define sib_label 0 "random pair" 1 "real pair"
label values siblingtype sib_label
```

```
bysort sibID: gen sib_n = _n
/*Double check that sib_n = 2!
/* Reshape to create new and unique sibIDs for all sibling pairs
reshape wide IBFlopNr sex FoddAr yrleavehome yrleave_14 whichparent sib_sex siblingtype, i(sibID)
j(sib_n)
gen sibID_new = _n
drop sibID
rename sibID_new = sibID
/* Reshape back to long format to add data
```

```
reshape long IBFLopNr sex FoddAr yrleavehome yrleave_14 whichparent sib_sex siblingtype sib_n,
i(sibID) j(sibID_within)
drop whichparent sib_n
```

```
save "$path\data.dta"
```

```
*****
*****Add data to chosen individuals*****
*****
```

```
*Basic ind data
```

```
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Basdata\IBF_IndData1990_2014.dta", keepusing(FoddAr sex Fland FlandFar)
drop if _merge == 2
drop _merge
```

```
*Recode country of birth of father
```

```
gen cob_father = 1 if FlandFar == 1
replace cob_father = 2 if FlandFar >= 2 & FlandFar <= 5
replace cob_father = 3 if FlandFar >= 6 & FlandFar <= 11
replace cob_father = 2 if FlandFar == 12
replace cob_father = 3 if FlandFar >= 13 & FlandFar <= 15
replace cob_father = 2 if FlandFar >= 16 & FlandFar <= 26
replace cob_father = 3 if FlandFar == 27
replace cob_father = 2 if FlandFar == 28
replace cob_father = 3 if FlandFar >= 29 & FlandFar <= 32
replace cob_father = 4 if FlandFar >= 33 & FlandFar <= 52
replace cob_father = 2 if FlandFar == 53
replace cob_father = 4 if FlandFar >= 54 & FlandFar <= 60
replace cob_father = 2 if FlandFar >= 61 & FlandFar <= 62
replace cob_father = 4 if FlandFar >= 63 & FlandFar <= 70
replace cob_father = 2 if FlandFar >= 71 & FlandFar <= 72
replace cob_father = 4 if FlandFar == 73
```

```
label variable cob_father "father's country of birth"
```

```
label define cob_labels 1 "Sweden" 2 "Western countries" 3 "Eastern Europe and Russia" 4 "Africa,
Asia, South America"
```

```
label values cob_father cob_labels
```

```
drop Fland FlandFar
```

```
*Calculate age
```

```
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
    gen age`i'=`i'-FoddAr
}
drop FoddAr
```

```
*Add neighbourhood (sams)
```

```
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000{
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(sams)
```

```

drop if _merge == 2
drop _merge
rename sams sams`i'
}

```

```

foreach i of numlist 2001 2003 2004 2005 2006 2007 2008 2009 2010{
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(SAMSomrade)
drop if _merge == 2
drop _merge
rename SAMSomrade sams`i'
}

```

```

*****
*****

```

```

/* Add LISA data
label define utb_label 1 "<12yrs" 2 "12yrs" 3 "13-14yr" 4 ">14yrs"
label define emp_label 1 "employed" 2 "not employed"
label define famtyp_label 1 "couple with children age <18" 2 "couple" 3 "single with children <18" 4
"single" 5 "other"
label define couple_label 1 "couple" 2 "single"
label define child_label 1 "children in household" 2 "no children in household"

```

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\ibf_loua`i'.dta", keepusing(Sun2000Niva SyssStatJ FamTypF)
drop if _merge == 2
drop _merge
recode Sun2000Niva (min/329 = 1) (330/399=2) (400/529=3) (nonmiss=4), gen(edu`i')
label values edu`i' utb_label
recode SyssStatJ (1 = 1) (nonmiss = 0), gen(emp`i')
label values emp`i' emp_label
recode FamTypF (12 22 = 1) (11 13 21 23 = 2) (31 41 = 3) (32 42 50 = 4) (nonmiss = 5), gen(famtype`i')
label values famtype`i' famtyp_label
recode famtype`i' (1/2 = 1) (3/5 = 2), gen(couple`i')
label values couple`i' couple_label
recode famtype`i' (1 3 = 1) (2 4 5 = 2), gen(child_hh`i')
label values child_hh`i' child_label
drop Sun2000Niva SyssStatJ FamTypF famtype`i'
}

```

```

*****
*****

```

```

/* Add income data - adjust for inflation and create logged variable
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004{

```

```

merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\förvärsinkomst_komp\förvärsinkomst`i'.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs forvers`i'
label variable forvers`i' "income from work`i'"
}

```

```

foreach i of numlist 2005 2006 2007 2008 2009 2010{
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\ibf_loua`i'.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs forvers`i'
label variable forvers`i' "income from work`i'"
}

```

*Adjust income of work for inflation - 1990 = 1

```

gen inc_KPI1990 = forvers1990
gen inc_KPI1991 = forvers1991/1.09
gen inc_KPI1992 = forvers1992/1.12
gen inc_KPI1993 = forvers1993/1.17
gen inc_KPI1994 = forvers1994/1.20
gen inc_KPI1995 = forvers1995/1.23
gen inc_KPI1996 = forvers1996/1.23
gen inc_KPI1997 = forvers1997/1.24
gen inc_KPI1998 = forvers1998/1.24
gen inc_KPI1999 = forvers1999/1.24
gen inc_KPI2000 = forvers2000/1.25
gen inc_KPI2001 = forvers2001/1.29
gen inc_KPI2002 = forvers2002/1.31
gen inc_KPI2003 = forvers2003/1.34
gen inc_KPI2004 = forvers2004/1.34
gen inc_KPI2005 = forvers2005/1.35
gen inc_KPI2006 = forvers2006/1.37
gen inc_KPI2007 = forvers2007/1.40
gen inc_KPI2008 = forvers2008/1.45
gen inc_KPI2009 = forvers2009/1.44
gen inc_KPI2010 = forvers2010/1.46

```

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
    replace inc_KPI`i' = 0.00001 if inc_KPI`i' == 0
    gen ln_inc`i'=ln(inc_KPI`i')
    drop inc_KPI`i' forvers`i'
}

```

```

*****
*****

```

```
/** Add tenure data
```

```
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003  
2004 2005 2006 2007{  
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Fastighet\beffast`i'.dta",  
keepusing(IBFFastLopNr)  
drop if _merge == 2  
drop _merge  
rename IBFFastLopNr IBFFastLopNr`i'  
}
```

```
foreach i of numlist 2001 2002 2003 2004 2005 2006 2007{  
rename IBFFastLopNr`i' IBFFastLopnr  
merge m:1 IBFFastLopnr using "D:\GeoSweden\GeoSweden Stata\Alla register\Fastighet\fast`i'.dta",  
keepusing(Typkod JurFormGrp)  
drop if _merge == 2  
drop _merge  
rename IBFFastLopnr IBFFastLopNr`i'  
rename Typkod typkod`i'  
rename JurFormGrp jurformgrp`i'  
}
```

```
label define segmentlabels 1 "home ownership" 2 "cooperative" 3 "rental"  
foreach i of numlist 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010{  
gen bostad`i'=0.  
recode typkod`i' (120/223=1) (320/325=2), gen(hustyp`i')  
replace bostad`i' = (hustyp`i'*10) + jurformgrp`i'  
recode bostad`i' (10/13=11) (16=1) (19=1) (17=2) (14/15=3) (18=4) (20/26=5) (29=5) (27=6) (28=7)  
recode bostad`i' (3=1) (2=2) (6=2) (1=3) (5=3) (4=3) (7=3), gen(tenure`i')  
label values tenure`i' segmentlabels  
replace tenure`i' = . if tenure`i' > 4  
drop bostad`i' typkod`i' hustyp`i' jurformgrp`i'  
}
```

```
save "$path\data.dta", replace
```

```
*****  
*****
```

```

*****
*****
*****Author: Lina Hedman [lina.hedman@ibf.uu.se]
*****Project: Siblings Neighbourhoods Effects
*****
*****
*****S L A V E (master: MASTER.do)*****
*****

*****Slave2_nbd.do
*****

```

```

/* Create neighbourhood files that identifies share low income earners

```

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000{
use "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\IBF_IndData1990_2014.dta",
keepusing(IBFLopNr FoddAr)
gen age`i' = `i'-FoddAr
gen workage if age`i' >= 20 & age`i' <= 64
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(sams)
drop if _merge == 2
drop _merge
rename sams sams`i'
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\förvärsinkomst_komp\förvärsinkomst`i'.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs forvers`i'
label variable forvers`i' "income from work`i'"
egen natdec`i'_low = pctlile(forvers`i') if workage == 1, p(30)
egen natdec`i'_high = pctlile(forvers`i') if workage == 1, p(70)
gen lowinc`i' = 1 if forvers`i' < natdec`i'_low & workage == 1
gen highinc`i' = 1 if forvers`i' > natdec`i'_high & workage == 1
collapse (count) inh_2064=IBFLopNr numlow=lowinc`i' numhigh=highinc`i' if workage == 1,
by(sams`i')
gen sh_low`i' = numlow/inh_2064*100
label variable sh_low`i' "share low income earners"
save "$path\sams`i'.dta"
}

```

```

foreach i of numlist 2001 2002 2003 2004{
use "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\IBF_IndData1990_2014.dta",
keepusing(IBFLopNr FoddAr)
gen age`i' = `i'-FoddAr
gen workage if age`i' >= 20 & age`i' <= 64
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(SAMSomrade)
drop if _merge == 2

```

```

drop _merge
rename SAMSomrade sams`i'
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\förväsinkomst_komp\förvärvsinkomst`i'.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs forvers`i'
label variable forvers`i' "income from work`i'"
egen natdec`i'_low = pctlile(forvers`i') if workage == 1, p(30)
egen natdec`i'_high = pctlile(forvers`i') if workage == 1, p(70)
gen lowinc`i' = 1 if forvers`i' < natdec`i'_low & workage == 1
gen highinc`i' = 1 if forvers`i' > natdec`i'_high & workage == 1
collapse (count) inh_2064=IBFLopNr numlow=lowinc`i' numhigh=highinc`i' if workage == 1,
by(sams`i')
gen sh_low`i' = numlow/inh_2064*100
label variable sh_low`i' "share low income earners"
save "$path\sams`i'.dta"
}

```

```

foreach i of numlist 2005 2006 2007 2008 2009 2010{
use "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\IBF_IndData1990_2014.dta",
keepusing(IBFLopNr FoddAr)
gen age`i' = `i'-FoddAr
gen workage if age`i' >= 20 & age`i' <= 64
merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(SAMSomrade)
drop if _merge == 2
drop _merge
rename SAMSomrade sams`i'
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\ibf_loua`i'.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs forvers`i'
label variable forvers`i' "income from work`i'"
egen natdec`i'_low = pctlile(forvers`i') if workage == 1, p(30)
egen natdec`i'_high = pctlile(forvers`i') if workage == 1, p(70)
gen lowinc`i' = 1 if forvers`i' < natdec`i'_low & workage == 1
gen highinc`i' = 1 if forvers`i' > natdec`i'_high & workage == 1
collapse (count) inh_2064=IBFLopNr numlow=lowinc`i' numhigh=highinc`i' if workage == 1,
by(sams`i')
gen sh_low`i' = numlow/inh_2064*100
label variable sh_low`i' "share low income earners"
save "$path\sams`i'.dta"
}

```

```
*****
*****Merge with individual file*****
*****
```

```
use "$path\data.dta", clear
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
merge m:1 sams`i' using "$path\sams`i'.dta", keepusing(sh_low`i')
drop if _merge == 2
drop _merge
}
save "$path\data.dta", replace
```

```
*****
*****
```

```

*****
*****
*****Author: Lina Hedman [lina.hedman@ibf.uu.se]
*****Project: Siblings Neighbourhood Effects
*****
*****
*****
***S L A V E (master: MASTER.do)*****
*****

```

```

*****
*****Slave3_randomsib.do
*****

```

```

*****
*****Generating random pairs within same neighbourhood and with same fathers birth country and
with same income level*****
*****

```

```

*****
*****Add basic data and select relent individuals*****
*****

```

```

use "G:\Data\Geosweden\GeoSweden STATA\Alla register\Basdata\inddata1990_2010.dta", clear
keep if FoddAr >= 1969 & FoddAr <= 1975
keep IBFLopNr FoddAr sex Fland FlandFar

```

```

/* Add family ID 1990 - 1997 for child and parents
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997{
    merge 1:1 IBFLopNr using "G:\Data\GeoSweden\GeoSweden STATA\Alla
register\Basdata\ibf_bef`i'.dta", keepusing(sams FamLopNr)
    drop if _merge == 2
    drop _merge
    rename sams sams`i'
    rename FamLopNr FamLopNr`i'
}

```

```

rename IBFLopNr IBFLopNr_child
rename IBFLopNrMor IBFLopNr

```

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997{
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
keepusing(FamLopNr)
drop if _merge == 2
drop _merge
rename FamLopNr FamLopNrMor`i'
}
rename IBFLopNr IBFLopNrMor
rename IBFLopNrFar IBFLopNr

```

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997{

```

```

merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef'i'.dta",
keepusing(FamLopNr)
drop if _merge == 2
drop _merge
rename FamLopNr FamLopNrFar`i'
}
/* Add income data of father for 1990
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\förvärsinkomst_komp\förvärsinkomst1990.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs ForvErs90_far
label variable ForvErs90_far "father's income from work 1990"

rename IBFLopNr IBFLopNrFar
rename IBFLopNr_child IBFLopNr

*Keep only those who live with parents in 1990
gen livepar = 1 if FamLopNr1990 == FamLopNrMor1990 | FamLopNr == FamLopNrFar1990
keep if livepar == 1
drop livepar

*year of leaving parental home - keep only those who move out between 1991 and 1997
gen yrleavehome = .
replace yrleavehome = 1991 if FamLopNr1990 == FamLopNrMor1990 | FamLopNr1990 ==
FamLopNrFar1990 & FamLopNr1990 != .
replace yrleavehome = 1992 if FamLopNr1991 == FamLopNrMor1991 | FamLopNr1991 ==
FamLopNrFar1991 & FamLopNr1991 != .
replace yrleavehome = 1993 if FamLopNr1992 == FamLopNrMor1992 | FamLopNr1992 ==
FamLopNrFar1992 & FamLopNr1992 != .
replace yrleavehome = 1994 if FamLopNr1993 == FamLopNrMor1993 | FamLopNr1993 ==
FamLopNrFar1993 & FamLopNr1993 != .
replace yrleavehome = 1995 if FamLopNr1994 == FamLopNrMor1994 | FamLopNr1994 ==
FamLopNrFar1994 & FamLopNr1994 != .
replace yrleavehome = 1996 if FamLopNr1995 == FamLopNrMor1995 | FamLopNr1995 ==
FamLopNrFar1995 & FamLopNr1995 != .
replace yrleavehome = 1997 if FamLopNr1996 == FamLopNrMor1996 | FamLopNr1996 ==
FamLopNrFar1996 & FamLopNr1996 != .
replace yrleavehome = 1998 if FamLopNr1997 == FamLopNrMor1997 | FamLopNr1997 ==
FamLopNrFar1997 & FamLopNr1997 != .
replace yrleavehome = . if yrleavehome == 1998 & FamLopNr1997 == .
replace yrleavehome = . if FamLopNr1990 == . & FamLopNr1991 == . & FamLopNr1992 == . &
FamLopNr1993 == . & FamLopNr1994 == . & FamLopNr1995 == . & FamLopNr1996 == . &
FamLopNr1997 == .
label variable yrleavehome "first year living away from parents"

keep if yrleavehome >= 1991 & yrleavehome <= 1997

gen yrleave_14 = yrleavehome+13
label variable yrleave_14 "year 14 years after having left parental home"

save "$path\random.dta"

```

```

/** Create income categories of father's income - using cutpoint identified in nbd data file
cut points income from national income distribution (identified when running nbd data)
gen inc_fa = 1 if ForvErs90_far < 878
replace inc_fa = 2 if ForvErs90_far >= 878 & ForvErs90_far < 1610
replace inc_fa = 3 if ForvErs90_far >= 1610
label variable inc_fa "income level of father"
label define inc_lab 1 "low inc" 2 "mid inc" 3 "high inc"
label values inc_fa inc_lab
drop ForvErs90_far

```

```

/** Country of birth of father
gen cob_father = 1 if FlandFar == 1
replace cob_father = 2 if FlandFar >= 2 & FlandFar <= 5
replace cob_father = 3 if FlandFar >= 6 & FlandFar <= 11
replace cob_father = 2 if FlandFar == 12
replace cob_father = 3 if FlandFar >= 13 & FlandFar <= 15
replace cob_father = 2 if FlandFar >= 16 & FlandFar <= 26
replace cob_father = 3 if FlandFar == 27
replace cob_father = 2 if FlandFar == 28
replace cob_father = 3 if FlandFar >= 29 & FlandFar <= 32
replace cob_father = 4 if FlandFar >= 33 & FlandFar <= 52
replace cob_father = 2 if FlandFar == 53
replace cob_father = 4 if FlandFar >= 54 & FlandFar <= 60
replace cob_father = 2 if FlandFar >= 61 & FlandFar <= 62
replace cob_father = 4 if FlandFar >= 63 & FlandFar <= 70
replace cob_father = 2 if FlandFar >= 71 & FlandFar <= 72
replace cob_father = 4 if FlandFar == 73

```

```

label variable cob_father "father's country of birth"
label values cob_father cob_labels
drop FlandFar Fland

```

```

*****
*****Create random sibling pairs*****
*****

```

```

generate random = runiform()
sort sams90 sex cob_father inc_fa random
by sams90 sex cob_father inc_fa: gen case_no = _n
by sams90 sex cob_father inc_fa: egen mean = mean(case_no)
gen sibno = 1 if case_no > mean
replace sibno = 2 if case_no <= mean
gen no1 = 1 if sibno == 1
gen no2 = 1 if sibno == 2
by sams90 sex cob_father inc_fa: gen case_id = sum(!missing(sibno)) if no1 == 1
by sams90 sex cob_father inc_fa: gen case_id2 = sum(!missing(sibno)) if no2 == 1
replace case_id = case_id2 if sibno == 2
tostring sams90, gen (s_string)
tostring sex, gen (sex_string)
tostring cob_father, gen (co_string)

```

```

tostring inc_fa, gen (i_string)
tostring case_id, gen (c_string)
gen sibID = s_string + sex_string + co_string + i_string + c_string
destring sibID, replace
bysort sibID: gen number = _n
drop if number > 2
drop c_string sex_string co_string i_string s_string mean no1 no2 case_id2 case_no sibno random
case_id
rename number sibID_within
sort sibID sibID_within

```

```

gen test = 0
replace test = 1 if sams90 == sams90[_n+1] & sibID_within == 1
replace test = 1 if sams90 == sams90[_n-1] & sibID_within == 2
drop if test == 0
drop test

```

```

*****
*****Clean file*****
*****

```

```

reshape wide IBFLopNr FoddAr sex sams90 cob_father IBFLopNrMor IBFLopNrFar whichparent
inc_fa, i(sibID) j(sibID_within)

```

```

*Keep pairs where at least on leaves home between 1991 and 1993
keep if (yrleavehome1 >= 1991 & <=1993) | (yrleavehome2 >= 1991 & <=1993)

```

```

*Keep only siblings who leave home min 4 years apart
gen leave_diff = abs(yrleavehome1 - yrleavehome2)
keep if leave_diff <=4

```

```

*Keep those in right age range
gen agediff = FoddAr1 - FoddAr2
replace agediff = abs(agediff)
keep if agediff <= 3

```

```

*Drop any actually related pairs
gen sibling = 1 if (IBFLopNrMor1 == IBFLopNrMor2) & (IBFLopNrFar1 == IBFLopNrFar2)
gen half_sibling = 1 if (IBFLopNrMor1 == IBFLopNrMor2) | (IBFLopNrFar1 == IBFLopNrFar2)
drop if sibling == 1
drop if half_sibling == 1

```

```

*Gen sex ID var
gen sib_sex = 1 if sex1 == 1 & sex2 == 1
replace sib_sex = 2 if sex1 == 2 & sex2 == 2
drop if sex1 != sex2
label values sib_sex sibsex_label

```

```

*Create new sibID variable
replace sibID = _n

```

```

reshape long
drop IBFLopNrMor IBFLopNrFar sams# cob_father inc_fa leave_diff age_diff sibling half_sibling
FamLopNr# FamLopNrMor# FamLopNrFar#

```

```

save "$path\random.dta", replace

```

```

*****
*****
*****Add data*****
*****

```

```

/* sams and family ID
foreach i of numlist 1991 1992 1993 1994 1995 1996 1997{
    merge 1:1 IBFLopNr using "G:\Data\GeoSweden\GeoSweden STATA\Alla
    register\Basdata\ibf_bef`i'.dta", keepusing(sams`i')
    drop if _merge == 2
    drop _merge
    rename FamLopNr FamLopNr`i'
}
foreach i of numlist 1998 1999 2000{
    merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
    keepusing(sams`i')
    drop if _merge == 2
    drop _merge
}

```

```

foreach i of numlist 2001 2003 2004 2005 2006 2007 2008 2009 2010{
    merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Basdata\ibf_bef`i'.dta",
    keepusing(SAMSomrade)
    drop if _merge == 2
    drop _merge
    rename SAMSomrade sams`i'
}

```

```

destring sams2009, replace force
destring sams2010, replace force

```

```

*****
*****

```

```

*Age
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
    gen age`i'=`i'-FoddAr
}
drop FoddAr

```

```

*****
*****

```

```

/* Add LISA data

```

```

label define utb_label 1 "<12yrs" 2 "12yrs" 3 "13-14yr" 4 ">14yrs"
label define emp_label 1 "employed" 2 "not employed"
label define famtyp_label 1 "couple with children age <18" 2 "couple" 3 "single with children <18" 4
"single" 5 "other"
label define couple_label 1 "couple" 2 "single"
label define child_label 1 "children in household" 2 "no children in household"

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\ibf_loua`i'.dta", keepusing(Sun2000Niva SyssStatJ FamTypF)
drop if _merge == 2
drop _merge
recode Sun2000Niva (min/329 = 1) (330/399=2) (400/529=3) (nonmiss=4), gen(edu`i')
label values edu`i' utb_label
recode SyssStatJ (1 = 1) (nonmiss = 0), gen(emp`i')
label values emp`i' emp_label
recode FamTypF (12 22 = 1) (11 13 21 23 = 2) (31 41 = 3) (32 42 50 = 4) (nonmiss = 5), gen(famtype`i')
label values famtype`i' famtyp_label
recode famtype`i' (1/2 = 1) (3/5 = 2), gen(couple`i')
label values couple`i' couple_label
recode famtype`i' (1 3 = 1) (2 4 5 = 2), gen(children`i')
label values children`i' child_label
drop Sun2000Niva SyssStatJ FamTypF famtype`i'
}

```

```

*****
*****

```

```

/* Add income data - adjust for inflation and create logged variable
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004{
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\förvärsinkomst_komp\förvärsinkomst`i'.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs forvers`i'
label variable forvers`i' "income from work`i'"
}

```

```

foreach i of numlist 2005 2006 2007 2008 2009 2010{
merge m:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla
register\Louise_Lisa\ibf_loua`i'.dta", keepusing(ForvErs)
drop if _merge == 2
drop _merge
rename ForvErs forvers`i'
label variable forvers`i' "income from work`i'"
}

```

*Adjust income of work for inflation - 1990 = 1

```
gen inc_KPI1990 = forvers1990
gen inc_KPI1991 = forvers1991/1.09
gen inc_KPI1992 = forvers1992/1.12
gen inc_KPI1993 = forvers1993/1.17
gen inc_KPI1994 = forvers1994/1.20
gen inc_KPI1995 = forvers1995/1.23
gen inc_KPI1996 = forvers1996/1.23
gen inc_KPI1997 = forvers1997/1.24
gen inc_KPI1998 = forvers1998/1.24
gen inc_KPI1999 = forvers1999/1.24
gen inc_KPI2000 = forvers2000/1.25
gen inc_KPI2001 = forvers2001/1.29
gen inc_KPI2002 = forvers2002/1.31
gen inc_KPI2003 = forvers2003/1.34
gen inc_KPI2004 = forvers2004/1.34
gen inc_KPI2005 = forvers2005/1.35
gen inc_KPI2006 = forvers2006/1.37
gen inc_KPI2007 = forvers2007/1.40
gen inc_KPI2008 = forvers2008/1.45
gen inc_KPI2009 = forvers2009/1.44
gen inc_KPI2010 = forvers2010/1.46
```

```
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
    replace inc_KPI`i' = 0.00001 if inc_KPI`i' == 0
    gen ln_inc`i'=ln(inc_KPI`i')
    drop inc_KPI`i'
    drop forvers
}
```

```
*****
*****
```

/* Add tenure data

```
foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007{
    merge 1:1 IBFLopNr using "D:\GeoSweden\GeoSweden Stata\Alla register\Fastighet\beffast`i'.dta",
    keepusing(IBFFastLopNr)
    drop if _merge == 2
    drop _merge
    rename IBFFastLopNr IBFFastLopNr`i'
}
```

```
foreach i of numlist 2001 2002 2003 2004 2005 2006 2007{
    rename IBFFastLopNr`i' IBFFastLopnr
```

```

merge m:1 IBFFastLopnr using "D:\GeoSweden\GeoSweden Stata\Alla register\Fastighet\fast`i'.dta",
keepusing(Typkod JurFormGrp)
drop if _merge == 2
drop _merge
rename IBFFastLopnr IBFFastLopNr`i'
rename Typkod typkod`i'
rename JurFormGrp jurformgrp`i'
}

```

```

label define segmentlabels 1 "home ownership" 2 "cooperative" 3 "rental"
foreach i of numlist 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010{
gen bostad`i'=0.
recode typkod`i' (120/223=1) (320/325=2), gen(hustyp`i')
replace bostad`i' = (hustyp`i'*10) + jurformgrp`i'
recode bostad`i' (10/13=11) (16=1) (19=1) (17=2) (14/15=3) (18=4) (20/26=5) (29=5) (27=6) (28=7)
recode bostad`i' (3=1) (2=2) (6=2) (1=3) (5=3) (4=3) (7=3), gen(tenure`i')
label values tenure`i' segmentlabels
replace tenure`i' = . if tenure`i' > 4
}

```

```

*****
*****Add neighbourhood data*****
*****

```

```

foreach i of numlist 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003
2004 2005 2006 2007 2008 2009 2010{
merge m:1 sams`i' using "$path\sams`i'.dta", keepusing(sh_low`i')
drop if _merge == 2
drop _merge
}

```

```

*****
*****Merging random and real siblings*****
*****

```

```

/* generate new identifier that is different from values used for real siblings
replace sibID = sibID*10000 if siblingtype == 0

```

```

/* generate variable to identify random pairs (same is in real sib file)
gen siblingtype = 0
label variable siblingtype "real or random pair"
label values siblingtype sib_label

```

```

save "$path\random.dta", replace

```

```

/* merge data
use "$path\data.dta", clear
append using "$path\random.dta"
/* drop any variables that do not appear in both files, if any
save "$path\data.dta", replace

```

```

*****
*****
*****Author: Lina Hedman [lina.hedman@ibf.uu.se]
*****Project: Siblings nbd effects
*****
*****
*****S L A V E (master: MASTER.do)*****
*****

```

```

*****
*****Slave4_analysis.do
*****

```

```

*****
*****
*****Transform into panel mode*****
*****

```

```

use "$path\data.dta", clear
reshape long ln_inc age edu emp couple child_hh tenure sams sh_low, i(IBFLopNr) j(year)

```

```

gen sams_parent = sams if year == yrleavehome - 1
bysort sibID sibID_within: egen sams_par = mean(sams_parent)
drop sams_parent

```

```

/* drop all years before leaving home and later than 14 years after. Create variable "years since
leave"
gen before = 1 if year < yrleavehome
gen after = 1 if year > yrleave_13
drop if before == 1
drop if after == 1
drop before after
bysort IBFLopNr year: egen yrsinceleave = _n
drop year
sort sibID sibID_within yrsinceleave

```

```

*****
*****Create new variables*****
*****

```

```

/* get income averaged over 3 years (12-14 yrs since leave)
gen inc_12 = ln_inc if yrsinceleave >= 12
bysort sibID sibID_within: egen inc_av = mean(inc_12)
drop inc_12
/* replace any negative incomes with zeros
recode inc_av (min/0 = 0)

```

```

/* get cumulative share low incom neighbours
sort sibID sibID_within yrsinceleave
gen shlow_cum = sh_low if yrsinceleave == 1

```

```
replace shlow_cum = shlow_cum[_n-1] + sh_low if sibID == sibID[_n-1] & sibID_within ==  
sibID_within[_n-1]
```

```
*****  
*****Only keep line of 11 years since leave (and remember to use inc_av, not ln_inc, as  
dependent!*****  
*****
```

```
keep if yrsinceleave == 11
```

```
/* generate family mean and ind departure variables for family model  
by sort sibID: egen fammean_shlowc = mean(shlow_cum)  
gen intrafammean_shlowc = shlow_cum - fammean_shlowc
```

```
save "$path\data.dta", replace
```

```
*****  
*****Clean file*****  
*****
```

```
/* delete if sams_par is not same for both siblings. Drop all in sibling pair  
drop if sams_par == .  
by sort sibID sibID_within: egen test_samspar = mean(sams_par)  
drop if sams_par != test_samspar  
drop test_samspar  
by sort sibID: egen test = count(sibID)  
drop if test == 1  
drop test
```

```
/* delete all cases with missing dep var and their siblings  
drop if inc_av == .  
by sort sibID: egen test = count(sibID)  
drop if test == 1  
drop test
```

```
/* delete all cases where present sams is missing and their siblings  
drop if sams == .  
by sort sibID: egen test = count(sibID)  
drop if test == 1  
drop test
```

```
/* generate new identifier variables  
sort sibID sibID_within  
gen indID = _n
```

```
sort sibID_within sibID  
gen famID = _n if sibID_within == 1  
sort sibID sibID_within  
replace famID = famID[_n-1] if sibID_within == 2 & sibID == sibID[_n-1]
```

```
save "$path\data.dta", replace
```

```
*****
*****
*****Descriptives*****
*****
*****
```

```
/* Run Decriptives
```

```
bysort siblingtype: summarize inc_av shlow_cum fammean_shlowc intrafammean_shlowc age if
sib_sex == 1
bysort siblingtype: summarize inc_av shlow_cum fammean_shlowc intrafammean_shlowc age if
sib_sex == 2
bysort siblingtype: tab cob_par if sib_sex == 1
bysort siblingtype: tab couple if sib_sex == 1
bysort siblingtype: tab child_hh if sib_sex == 1
bysort siblingtype: tab edu if sib_sex == 1
bysort siblingtype: tab emp if sib_sex == 1
bysort siblingtype: tab tenure if sib_sex == 1
bysort siblingtype: tab cob_father if sib_sex == 2
bysort siblingtype: tab couple if sib_sex == 2
bysort siblingtype: tab child_hh if sib_sex == 2
bysort siblingtype: tab edu if sib_sex == 2
bysort siblingtype: tab emp if sib_sex == 2
bysort siblingtype: tab tenure if sib_sex == 2
```

```
*****
*****
*****Models*****
*****
*****
```

```
*****
*****Males*****
*****
```

```
/* Real siblings
```

```
xi:mixed inc_av shlow_cum age i.cob_father if siblingtype == 1 & sib_sex == 1 || sams_par:
|| famID:, mle var
```

```
xi:mixed inc_av shlow_cum age i.cob_father i.couple i.child_hh i.edu i.emp i.tenure if siblingtype ==
1 & sib_sex == 1 || sams_par: || famID:, mle var
```

```
xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father if siblingtype == 1 &
sib_sex == 1 || sams_par: || famID:, mle var
```

```
xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father i.couple i.child_hh i.edu
i.emp i.tenure if siblingtype == 1 & sib_sex == 1 || sams_par: || famID:, mle var
```

```

/**Random siblings
xi:mixed inc_av shlow_cum age i.cob_father if siblingtype==0 & sib_sex == 1 || sams_par: || famID;, mle var

xi:mixed inc_av shlow_cum age i.cob_father i.couple i.child_hh i.edu i.emp i.tenure if siblingtype==0
& sib_sex == 1 || sams_par: || famID;, mle var

xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father if siblingtype==0 & sib_sex
== 1 || sams_par: || famID;, mle var

xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father i.couple i.child_hh i.edu
i.emp i.tenure if siblingtype==0 & sib_sex == 1 || sams_par: || famID;, mle var

*****
***Females*****
*****

/** Real siblings
xi:mixed inc_av shlow_cum age i.cob_father if siblingtype == 1 & sib_sex == 2 || sams_par: ||
famID;, mle var

xi:mixed inc_av shlow_cum age i.cob_father i.couple i.child_hh i.edu i.emp i.tenure if siblingtype == 1
& sib_sex == 2 || sams_par: || famID;, mle var

xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father if siblingtype == 1 &
sib_sex == 2 || sams_par: || famID;, mle var

xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father i.couple i.child_hh i.edu
i.emp i.tenure if siblingtype == 1 & sib_sex == 2 || sams_par: || famID;, mle var

/**Random siblings
xi:mixed inc_av shlow_cum age i.cob_father if siblingtype==0 & sib_sex == 2 || sams_par: || famID;,
mle var

xi:mixed inc_av shlow_cum age i.cob_father i.couple i.child_hh i.edu i.emp i.tenure if siblingtype==0
& sib_sex == 2 || sams_par: || famID;, mle var

xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father if siblingtype==0 & sib_sex
== 2 || sams_par: || famID;, mle var

xi:mixed inc_av fammean_shlowc intrafammean_shlowc age i.cob_father i.couple i.child_hh i.edu
i.emp i.tenure if siblingtype==0 & sib_sex == 2 || sams_par: || famID;, mle var

```